Code No: 183AM **R22**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech II Year I Semester Examinations, February - 2024

DATA STRUCTURES

(Common to CSE(AI&ML), CSE(IOT), AI&DS, AI&ML)

Time: 3 Hours **Max. Marks: 60**

Note: This question paper contains two parts A and B.
i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

PART- A

(10 Marks)

1.a)	Define a single linked list?	[1]
b)	What are the applications of the queue?	[1]
c)	Define Rehashing?	[1]
d)	What is a skip list?	[1]
e)	Define Binary search tree?	[1]
f)	What is the need of rotations in red-black tree?	[1]
g)	List out the external sorting techniques.	[1]
h)	Define a graph.	[1]
i)	Define a trie.	[1]
j)	List out the applications of tries.	[1]

PART-B

(50 Marks)

2.a) Write an algorithm for Insertion and Deletion operations in a Singly Linked List.
b) Explain Push and Pop operations of a stack using array representations. [5+5]

OR

3.a) The following numbers 10, 20, 50, 30, 90, 60 (Top) are present in a stack of size 10. Perform the following operations in sequence. pop(), push(30), push(40), pop(), push (60), pop(), pop(), pop() What is the peek element at last? Draw and explain it?
b) Convert the following infix expression into postfix expression:
A + B – C * D * E / F – G. [5+5]

4.a) What is a Dictionary?
b) Write an algorithm for insertion and deletion operations of a skip list. [5+5]

OR

5.a) Insert the following list of elements into the hash table by using Quadratic probing (size of Hash table is 13: 65, 34, 79, 114, 26, 85, 55, 89, 22, 98.
b) Explain briefly about double hashing and extendible hashing techniques. [4+6]

6.a) Write an algorithm to delete an element from the binary search tree and provide an example.

b) In an initially empty AVL tree insert the following keys: DEC, JAN, APR,MAR, JUL, AUG, OCT, FEB, NOV. Draw AVL tree after every insertion and apply rotations wherever is necessary? [5+5]

OR

7.a) Discuss the importance of height balanced trees for searching.

b) Insert the following list of elements from the Red- Black tree 12, 30, 36, 18, 25, 9, 4, 2, 17, 14, 20, 47. Delete the elements 18, 2and 30 from the Red-Black tree and draw the final tree diagram. [5+5]

8.a) Consider five cities: (i) New Delhi, (ii) Mumbai, (iii) Chennai, (iv) Bangalore, and (v) Kolkata, and a list of flights that connect these cities as shown in the following table. Use the given information to construct a graph? Also represent the graph using adjacency matrix and adjacency list

Flight No	Origin	Destination
101	2	3
102	3	2
103	5	3
104	3	4
105	2	5
106	5	2
107	5	1
108	1	4
109	5	4
110	4	5

b) Write and explain depth-first search traversal algorithm with an example. [5+5]

OR

9.a) Write and perform heap sort algorithm for (10 15 6 2 25 18 16 2 20 4) .

b) Explain BFS algorithm with example. [5+5]

10.a) Write an algorithm of Standard Trie.

b) Solve the Knuth Morris-Pratt algorithm for the following:
Example:Text: HEREISASIMPLEEXAMPLE
Pattern: EXAMPLE. [5+5]

OR

11.a) "A compressed trie is an advanced version of the standard trie." Support or oppose this statement with necessary explanation.

b) Explain the features that distinguish between Boyer Moore algorithm from the conventional algorithms. [5+5]

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